

Executive Proposal: Standardizing Media Metadata Extraction via In-House Engine

Status: Proposal & Validated POC

Target System: The Vault Media Pipeline

Date: August 2026

1. Executive Summary

Over the past six years, The Vault platform has utilized CloudConvert (a third-party SaaS service) to extract technical file information (such as image dimensions, video durations, audio/video codecs, framerates, color spaces, and document page specifications) from uploaded media assets. This information is saved directly to our database and powers downstream processes such as asset validation, automated layout rendering, and media delivery.

Operating an external third-party service for metadata extraction over an extended period has introduced key operational challenges:

- **Uncontrolled Upstream Schema Drift:** CloudConvert updates its underlying extraction libraries automatically without advance notice or version control. These updates have introduced subtle changes to metadata property names over time, requiring our engineering team to continually write complex backwards-compatibility code (such as checking multiple variations of dimensions like ImageWidth, width, ImageSize, and MaxPageSizeW) to ensure older campaign assets do not break modern rendering workflows.
- **Operational Latency & Webhook Dependency:** External API roundtrips and remote queuing routinely add between 2 and 28 seconds of latency per asset, keeping Lambda functions alive longer and introducing an external network dependency into core ingestion.
- **Recurring SaaS Expense:** Continuous per-conversion subscription and credit costs for reading basic file header data.

Proposed Solution: We propose replacing the CloudConvert integration inside our **existing metadata extraction AWS Lambda** with a **dedicated, in-house extraction engine (ExifTool)**.

Because the Lambda infrastructure is already provisioned and executes asynchronously, infrastructure compute costs remain virtually unchanged, while completely eliminating external SaaS billing. To further improve resilience, we recommend introducing a standard message queue (Amazon SQS) between the backend and Lambda to buffer high-volume upload bursts.

2. Ingestion Architecture & Technical Rationale

Current Architecture Workflow

CURRENT CLOUDCONVERT SAAS WORKFLOW

```
1. [User / Frontend] —(Direct File Upload)—> [Amazon S3] 2. [Frontend] —(Upload Complete API Call)—> [Backend Service] 3. [Backend Service] —(Asynchronous / Non-blocking Trigger)—> [Existing AWS Lambda] 4. [Existing AWS Lambda] —(External API Call: 2s - 28s)—> [CloudConvert SaaS] 5. [Existing AWS Lambda] —(Direct Database Save)—> [PostgreSQL files.metadata]
```

- **Step 1:** The user uploads the media file directly from the browser to Amazon S3.
- **Step 2:** Once the S3 upload finishes, the frontend notifies the backend via an API call.
- **Step 3:** The backend immediately triggers the metadata extraction AWS Lambda in an **asynchronous, non-blocking (fire-and-forget)** manner, allowing the API call to complete instantly without waiting for metadata extraction.
- **Step 4:** The Lambda communicates with the external CloudConvert API to process the file and retrieve technical headers.
- **Step 5:** When extraction completes, the **Lambda writes the metadata directly into the PostgreSQL database** using its direct database connection.

Target Architecture (In-House Engine with Queue Buffering)

TARGET IN-HOUSE EXIFT00L WORKFLOW

```
1. [User / Frontend] —(Direct File Upload)—> [Amazon S3] 2. [Frontend] —(Upload Complete API Call)—> [Backend Service] 3. [Backend Service] —(Enqueue Task)—> [Amazon SQS Queue Buffer] 4. [Amazon SQS Queue] —(Asynchronous Trigger)—> [Existing AWS Lambda (In-House Engine: <400ms)] 5. [Existing AWS Lambda] —(Direct Database Save)—> [PostgreSQL files.metadata]
```

Key Problem Analyses & Solutions

- **Problem 1 (Upstream Version Changes & Data Inconsistencies):** Because CloudConvert is an unversioned third-party service, updates to its internal toolchain occur without notice. Over six years of operation, this has resulted in mixed data structures across historical campaign files (older assets contain legacy attribute names; newer assets contain updated names).

The Solution: By deploying the engine directly within our existing Lambda, we lock in a specific, stable version. The data structure remains 100% consistent and predictable. Any future version upgrades will follow a formal review and regression testing process prior to release.

- **Problem 2 (Extraction Latency & Extended Lambda Lifecycles):** Currently, when the Lambda runs, it must wait for CloudConvert to create an external job, queue the task on remote servers, process it, and respond over the public internet (routinely taking 2 to 28 seconds per asset).
The Solution: With the in-house engine embedded directly in the Lambda, the function reads the file header from S3 and completes extraction in **under 400 milliseconds**, then immediately updates the database. This reduces Lambda execution time by over 90% and eliminates external API failure risks.
- **Problem 3 (Zero Additional Infrastructure Cost):** Because an AWS Lambda function is already in place to handle metadata triggers and database writes, this upgrade simply updates the logic inside the existing function. There is no need to purchase new compute instances or increase infrastructure spend, while 100% of third-party CloudConvert credit fees are eliminated.

3. Proof of Concept (POC) Validation & Benchmark Results

A Proof of Concept was developed to verify data parity between live CloudConvert API responses and the in-house extraction engine.

A. Testing Scope Across Unique Media Formats

Validation was conducted against representative sample assets covering all core media formats supported by The Vault:

MEDIA FORMAT	ASSET TYPE	EXTRACTED TECHNICAL ATTRIBUTES	VERIFICATION STATUS
PNG	Brand graphics, transparent overlays	Dimensions, Bit Depth, Color Type, DPI, Gamma	100% Parity
JPEG	Photography, campaign images	Dimensions, EXIF metadata, Megapixels, Color Space	100% Parity
MP4	Video commercials, board renders	Resolution, Duration, Framerate, Bitrate, Audio/Video Codecs	100% Parity
QuickTime MOV	High-definition master videos	Resolution, TimeScale, Duration, Codec profile	100% Parity
PDF	Print proofs, campaign specifications	Page Count, PDF Version, Creator Software, Linearization	100% Parity
Adobe Illustrator (AI / EPS)	Vector artwork, signage dimensions	Artboard Dimensions, Bounding Box, XMP metadata	100% Parity
Adobe Photoshop (PSD)	Multi-layer master graphics	Layer Count, Blend Modes, Layer Names, Color Profile	100% Parity
WebP	Web banners	Dimensions, Compression mode, Alpha channel	100% Parity

B. Live Real-Time Benchmark Comparison

A live benchmark was conducted comparing real-time CloudConvert API responses against the in-house engine on identical assets:

BENCHMARK DIMENSION	CLOUDCONVERT SAAS (LIVE API)	IN-HOUSE ENGINE (PROPOSED)	OPERATIONAL ADVANTAGE
Processing Latency	~28,990 ms (Upload + Queue + Network)	381 ms (Direct stream parse)	76x Faster Extraction
Critical Field Match Rate	100%	100%	Exact Match
Width & Height Accuracy	Identical	Identical	100% Match
File Format & MIME Type	Identical	Identical	100% Match
Color Profiling & Bit Depth	Identical	Identical	100% Match
Third-Party SaaS Cost	Recurring monthly credit billing	\$0.00	Complete Cost Elimination

Technical Confirmation: The outputs match identically because CloudConvert uses this exact open-source engine internally. Running it directly within our existing AWS Lambda delivers the exact same data without third-party transit or recurring subscription fees.

4. Architectural Enhancements & Stability

In addition to replacing the external service call inside the Lambda, we recommend adding an **Amazon SQS message queue buffer** between the backend API and the Lambda:

- **Burst Upload Protection:** During large campaign launches where multiple assets are uploaded simultaneously, the queue cleanly buffers incoming jobs. This ensures the Lambda processes files steadily without spiking concurrent database connections.
- **Automated Error Handling:** If an invalid or corrupted file is uploaded, the queue safely routes it to a Dead-Letter Queue (DLQ) for alerting, without interrupting the processing of other assets.
- **Enhanced Security:** Media assets remain entirely inside our private AWS environment and are never transmitted to external third-party infrastructure.

5. Rollout Strategy

- Phase 1 (Staging Verification):** Deploy the updated Lambda function with the in-house engine to staging environments for end-to-end workflow validation.
- Phase 2 (Dual Verification in Production):** Run the in-house extraction alongside the existing process for a 14-day observation window to confirm 100% parity across live production uploads.
- Phase 3 (Full Cutover & Decommissioning):** Decommission the external CloudConvert integration and terminate third-party credit subscriptions.

6. Strategic Business Impact Summary

STRATEGIC AREA	CURRENT STATE (CLOUDCONVERT SAAS)	PROPOSED STATE (IN-HOUSE ENGINE)	BUSINESS BENEFIT
Third-Party SaaS Cost	Recurring monthly credit billing	\$0.00 / month	Direct operational savings.
Infrastructure Cost	Existing Lambda	Existing Lambda (No change)	Zero added compute spend.
Processing Speed	2 to 28 seconds per asset	Under 400 milliseconds	Snappier UI and faster asset availability.
Data Consistency	Vulnerable to unannounced upstream changes	Fixed, version-controlled schema	Eliminates code complexity and maintenance overhead.
Queue Stability	Direct invocation model	Message queue buffer (Amazon SQS)	Higher reliability during peak upload volumes.
Data Privacy	Files sent to external third-party servers	Contained within private AWS cloud	Full data privacy and enterprise compliance.

7. Conclusion

Replacing the external CloudConvert integration with an in-house engine inside our existing Lambda resolves the long-standing schema drift challenge, accelerates asset ingestion from multi-second queues to sub-second execution, eliminates recurring SaaS expenses, and introduces message queue buffering for peak upload stability.

